EXPERIMENT 5: Write a C program for generalization of the Caesar cipher, known as the affine Caesar cipher, has the following form: For each plaintext letter p, substitute the ciphertext letter C: C = E([a, b], p) = (ap + b)

CODE:

```
#include <stdio.h>

#include <string.h>

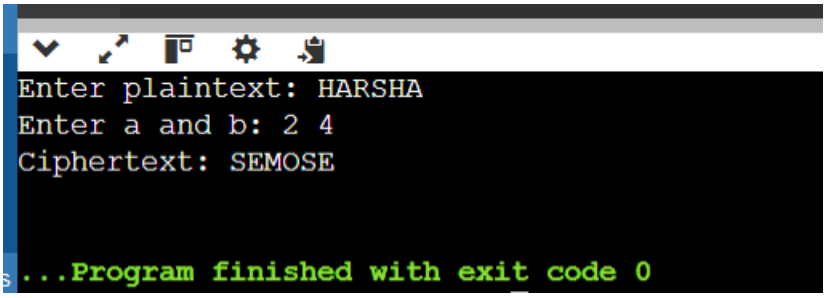

int main() {
   char s[100];
   int a, b, i;


   printf("Enter plaintext: ");
   scanf("%99s", s);
   printf("Enter a and b: ");
   scanf("%d%d", &a, &b);


   for (i = 0; s[i]; i++) {
      if (s[i] >= 'A' && s[i] <= 'Z')
         s[i] = (a * (s[i] - 'A') + b) % 26 + 'A';
      else if (s[i] >= 'a' && s[i] <= 'z')
         s[i] = (a * (s[i] - 'a') + b) % 26 + 'a';
   }


   printf("Ciphertext: %s\n", s);
   return 0;
}
```

OUTPUT:


Enter plaintext: HARSHA
Enter a and b: 2 4
Ciphertext: SEMOSE
...Program finished with exit code 0